

4. Basics of Probabilistic Modelling

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Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Statistical Estimation and Hypothesis Testing
4. Basics of Probabilistic Modelling

M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Lecture Topics

- ▶ We have previously formulated Bayesian networks and explored statistical inference based on them.
- ▶ In this chapter, we begin by reviewing two fundamental paradigms in probabilistic modeling: discriminative and generative models.
- ▶ After discussing their limitations in high-dimensional settings, we will transition into traditional dimension reduction methods.
- ▶ Recommended reference materials include:
 - ▶ Efron (1975), "The Efficiency of Logistic Regression to Normal Discriminant Analysis," *JASA*.
 - ▶ Section 14 in Hastie (2009), *The Elements of Statistical Learning*.

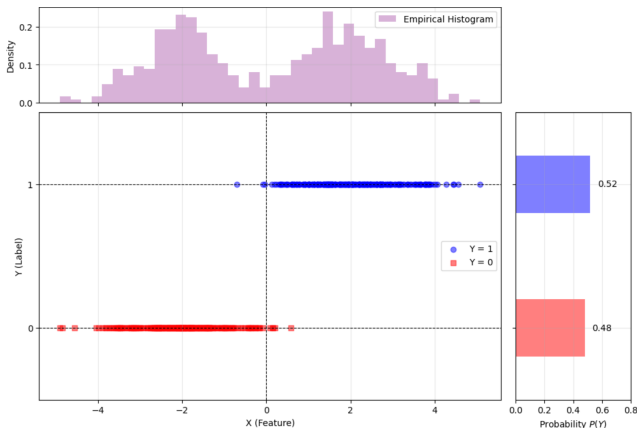
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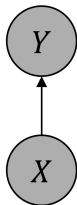
Discriminative vs. Generative Models



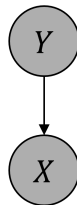
- ▶ Given data, there are multiple ways to describe the dependency structure.
- ▶ For example, let X be an explanatory variable and Y be a binary response.
- ▶ Q: How can we construct a probabilistic model for $(X, Y)^T$ and learn $p(y | x)$?^a

^aThis task is referred to as *binary classification*.

Discriminative vs. Generative Models



(a) Discriminative Model



(b) Generative Model

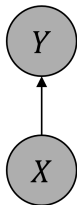
► We may consider two approaches:

1. **Discriminative Model**: Models only the conditional distribution $p(y|x)$.
2. **Generative Model**: Models the joint distribution

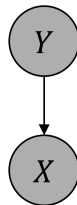
$$f(x, y) = p(y)f(x | y), \tag{1}$$

typically by modelling $p(y)$ and $f(x|y)$ separately.

Discriminative Model



(a) Discriminative Model



(b) Generative Model

- **Discriminative Model**: We consider *logistic regression*. First, we may consider:

$$X \sim f(x) \quad \text{and} \quad Y \mid X = x \sim \text{Ber}(\mu_\theta(x)), \quad (2)$$

where $f(x)$ does not depend on the parameter θ , and $\mu_\theta(x)$ is a parameterized conditional success probability designed to model the dependence of Y on X .

Discriminative Model

- ▶ In this case, for a given x , the conditional log-PMF is expressed as:

$$\begin{aligned}\log p_{\theta}(y | x) &= \log \text{Ber}(y; \mu_{\theta}(x)) \\ &= \log \left(\mu_{\theta}(x)^y (1 - \mu_{\theta}(x))^{1-y} \right) \\ &= y \log \frac{\mu_{\theta}(x)}{1 - \mu_{\theta}(x)} + \log(1 - \mu_{\theta}(x)).\end{aligned}\tag{3}$$

- ▶ Note that $\log p_{\theta}(y | x)$ is a linear function of y with the coefficient:

$$\eta_{\theta}(x) = \log \frac{\mu_{\theta}(x)}{1 - \mu_{\theta}(x)},^1\tag{4}$$

which is the *logit* (or *log-odds*) of success given x .

¹It is formally referred to as the *natural parameter* in the context of the *exponential family*.

Discriminative Model

- ▶ Here, letting $\text{sig}(u) := 1/(1 + \exp(-u))$ denote the *sigmoid function* (or *logistic function*), we have:

$$\mu_{\theta}(x) = \text{sig}(\eta_{\theta}(x)) = \frac{1}{1 + \exp(-\eta_{\theta}(x))}.^2 \quad (5)$$

- ▶ For the choice of $\eta_{\theta}(\cdot)$, we consider the *logistic model* where the logit $\eta_{\theta}(x)$ is a linear function of x :

$$\left\{ \eta_{\theta}(x) = \beta_0 + \beta_1 x \mid \theta = (\beta_0, \beta_1)^T \in \mathbb{R}^2 \right\}. \quad (6)$$

- ▶ Finding the MLE within this model class leads to *logistic regression*.

²In a similar manner, this formulation can be extended to multi-class classification using the *softmax function*.

Discriminative Model

- ▶ Let's formulate the likelihood maximization. The (joint) log-likelihood function can be expressed as:

$$\begin{aligned}\ell(\theta) &= \sum_{i=1}^n \log f_{\theta}(x_i, y_i) \\ &= \sum_{i=1}^n \log f(x_i) + \sum_{i=1}^n \left(y_i \eta_{\theta}(x_i) + \log(1 - \text{sig}(\eta_{\theta}(x_i))) \right).\end{aligned}\tag{7}$$

- ▶ Differentiating both sides with respect to θ yields:

$$\nabla_{\theta} \ell(\theta) = \sum_{i=1}^n (y_i - \text{sig}(\eta_{\theta}(x_i))) \nabla_{\theta} \eta_{\theta}(x_i).\tag{8}$$

Discriminative Model

- ▶ Differentiating both sides once more yields the *Hessian matrix*:

$$\nabla_{\theta}^2 \ell(\theta) = - \sum_{i=1}^n \text{sig}(\eta_{\theta}(x_i)) (1 - \text{sig}(\eta_{\theta}(x_i))) \nabla_{\theta} \eta_{\theta}(x_i) \nabla_{\theta} \eta_{\theta}(x_i)^T. \quad (9)$$

- ▶ For the logistic model class where $\eta_{\theta}(x) = \beta_0 + \beta_1 x$, we have:

$$\nabla_{\theta} \eta_{\theta}(x_i) = (1, x_i)^T. \quad (10)$$

Consequently, the Hessian $\nabla_{\theta}^2 \ell(\theta)$ becomes strictly negative definite provided that the sample variance satisfies $S_X^2 > 0$.

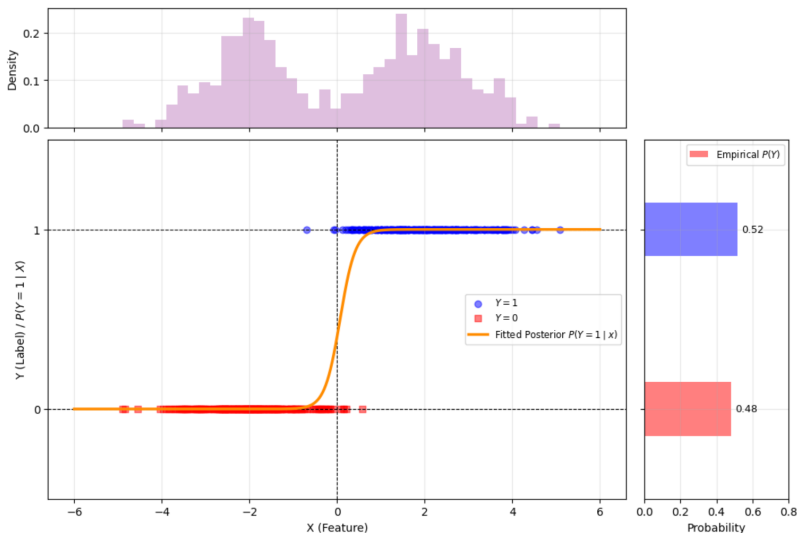
Discriminative Model

- ▶ Thus, $\ell(\theta)$ is strictly concave, and it is sufficient to solve the *score equation*:

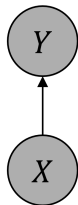
$$\begin{aligned} \nabla_{\theta} \ell(\theta) &= 0, \quad \text{or equivalently,} \\ \sum_{i=1}^n (y_i - \mu_{\theta}(x_i)) \begin{pmatrix} 1 \\ x_i \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \end{aligned} \tag{11}$$

- ▶ We can apply Newton's method, an iterative optimization algorithm, to numerically approximate the MLE.
- ▶ Introducing $\hat{p}_i := \mu_{\theta}(x_i) = P_{\theta}(Y = 1 \mid X = x_i)$, we have:

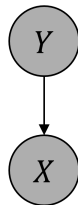
Discriminative Model



Generative Model



(a) Discriminative Model



(b) Generative Model

- **Generative Model**: We consider *linear discriminant analysis (LDA)*. First, we model the joint distribution directly:

$$Y \sim \text{Ber}(p) \quad \text{and} \quad X \mid Y = y \sim \mathcal{N}(\mu_y, \sigma^2), \quad (12)$$

where $p \in (0, 1)$ is the class prior, and the conditional distribution of X given Y is modeled as a class-specific Gaussian with a shared variance σ^2 .

Generative Model

- ▶ Let $\theta = (p, \mu_0, \mu_1, \sigma^2)^T$ be the parameter vector. Then, the joint density-PMF can be expressed as:

$$f_{\theta}(x, y) = \underbrace{\left(p^y (1-p)^{1-y} \right)}_{=p_{\theta}(y)} \underbrace{\left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(x - \mu_y)^2}{2\sigma^2} \right) \right)}_{=f_{\theta}(x|y)}. \quad (13)$$

- ▶ Consequently, by Bayes' rule, the posterior probabilities are given by:

$$P_{\theta}(Y = 1 \mid X = x) = \frac{f_{\theta}(x \mid y = 1)p_{\theta}(y = 1)}{f_{\theta}(x \mid y = 1)p_{\theta}(y = 1) + f_{\theta}(x \mid y = 0)p_{\theta}(y = 0)} \quad (14)$$

and

$$P_{\theta}(Y = 0 \mid X = x) = \frac{f_{\theta}(x \mid y = 0)p_{\theta}(y = 0)}{f_{\theta}(x \mid y = 1)p_{\theta}(y = 1) + f_{\theta}(x \mid y = 0)p_{\theta}(y = 0)}. \quad (15)$$

Generative Model

- ▶ Thus, given θ , the corresponding logit can be expressed as:

$$\eta_{\theta}(x) := \log \frac{P_{\theta}(Y = 1 \mid X = x)}{P_{\theta}(Y = 0 \mid X = x)}. \quad (16)$$

Generative Model

- ▶ Consequently, the decision boundary of this model, defined as the set of points where $\eta_{\theta}(x) = 0$, is strictly *linear* with respect to x :

$$\beta_0 + \beta_1 x = 0, \quad (17)$$

where we define the mapped parameters as:

$$\beta_0 := \log \frac{p}{1-p} - \frac{\mu_1^2 - \mu_0^2}{2\sigma^2} \quad \text{and} \quad \beta_1 := \frac{\mu_1 - \mu_0}{\sigma^2}. \quad (18)$$

- ▶ This demonstrates that the LDA naturally induces a logistic regression structure. We can thus use its MLE to calculate the predicted posterior probabilities.
- ▶ Conversely, the LDA can represent any (non-trivial) logistic model. When $\beta_1 \neq 0$,

$$(p, \mu_0, \mu_1, \sigma^2)^T = \left(0.5, -\frac{\beta_0}{\beta_1} - \frac{\beta_1}{2}, -\frac{\beta_0}{\beta_1} + \frac{\beta_1}{2}, 1 \right)^T \quad (19)$$

recovers the corresponding logistic model with $(\beta_0, \beta_1)^T$.

Generative Model

- Let's formulate the likelihood maximization. The log-likelihood function is given by:

$$\begin{aligned}\ell(\theta) &= \sum_{i=1}^n \log \left(p^{y_i} (1-p)^{1-y_i} \right) + \sum_{i=1}^n \log \left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(x_i - \mu_{y_i})^2}{2\sigma^2} \right) \right) \\ &= \sum_{i=1}^n \left(y_i \log p + (1-y_i) \log(1-p) - \frac{1}{2} \log(2\pi\sigma^2) - \frac{(x_i - \mu_{y_i})^2}{2\sigma^2} \right).\end{aligned}\tag{20}$$

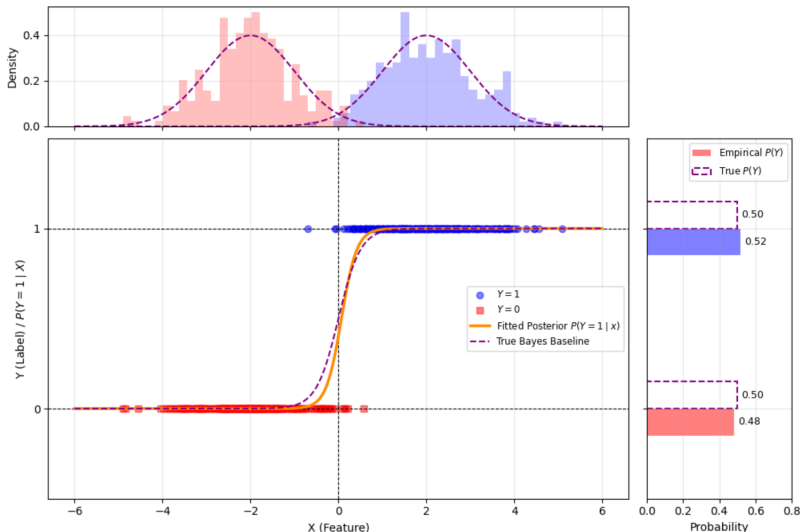
- Unlike logistic regression, $\ell(\theta)$ can be optimized analytically in closed form:

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n y_i, \quad \hat{\mu}_1 = \frac{\sum_{i=1}^n y_i x_i}{\sum_{i=1}^n y_i}, \quad \hat{\mu}_0 = \frac{\sum_{i=1}^n (1-y_i) x_i}{\sum_{i=1}^n (1-y_i)}, \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu}_{y_i})^2.\tag{21}$$

Generative Model

$$\blacktriangleright \ell(\theta) = \sum_{i=1}^n \left(y_i \log p + (1 - y_i) \log(1 - p) - \frac{1}{2} \log(2\pi\sigma^2) - \frac{(x_i - \mu_{y_i})^2}{2\sigma^2} \right).$$

Generative Model



Generative Model: Synthetic Data Generation

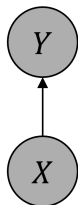
- Given the LDA generative model $f_{\theta}(x, y) = p_{\theta}(y)f_{\theta}(x | y)$, we can *generate* synthetic data as follows:
1. Draw a new label $y_{\text{new}} \sim \text{Ber}(p)$.
 2. Generate a uniform realization $u \sim \mathcal{U}(0, 1)$.
 3. Calculate x_{new} :

$$x_{\text{new}} = \mu_{y_{\text{new}}} + \sigma \Phi^{-1}(u). \quad (22)$$

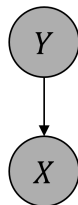
- More generally, for any class-conditional distribution with a known CDF $F_{X|Y=y}(x)$, we can draw $u \sim \mathcal{U}(0, 1)$ and compute:

$$x_{\text{new}} = F_{X|Y=y_{\text{new}}}^{-1}(u). \quad (23)$$

Discriminative vs. Generative Models



(a) Discriminative Model



(b) Generative Model

- Q: What would be the pros and cons of the two approaches?

(a) **Logistic Regression**: Directly models the posterior probability:

$$P_{\theta}(Y = 1 \mid X = x) = \text{sig}(\beta_0 + \beta_1 x). \quad (24)$$

(b) **Linear Discriminant Analysis**: Models the entire data-generating process:

$$f_{\theta}(x, y) = p_{\theta}(y)f_{\theta}(x \mid y), \quad (25)$$

where $Y \sim \text{Ber}(p)$, $X \mid Y = y \sim \mathcal{N}(\mu_y, \sigma^2)$, and $\theta = (p, \mu_0, \mu_1, \sigma^2)^T$.

Discriminative vs. Generative Models

- ▶ In the previous toy data, the true data-generating mechanism was as follows:

$$Y \sim \text{Ber}(p) \quad \text{and} \quad X \mid Y = y \sim \mathcal{N}(\mu_y, \sigma^2) \quad (26)$$

with $(p, \mu_0, \mu_1, \sigma^2)^T = (0.5, -2, 2, 1)^T$, implying $(\beta_0, \beta_1)^T = (0, 4)^T$. Both models correctly specify the true linear logit structure.

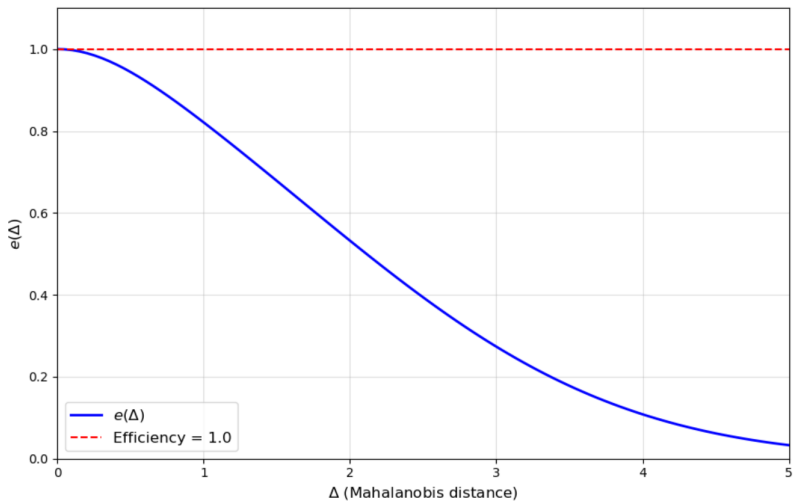
- ▶ When $p = 0.5$, the relative asymptotic efficiency for β_1 is expressed as (Efron, 1975):

$$e(\Delta) := \lim_{n \rightarrow \infty} \frac{\text{Var}(\hat{\beta}_{1,\text{LDA}})}{\text{Var}(\hat{\beta}_{1,\text{LR}})} = \frac{4 + \Delta^2}{2} \int_{-\infty}^{\infty} z^2 \frac{\phi\left(z - \frac{\Delta}{2}\right) \phi\left(z + \frac{\Delta}{2}\right)}{\phi\left(z - \frac{\Delta}{2}\right) + \phi\left(z + \frac{\Delta}{2}\right)} dz, \quad (27)$$

where $\Delta = \frac{|\mu_1 - \mu_0|}{\sigma}$ represents the *Mahalanobis distance*.

- ▶ In our specific setup ($\Delta = 4$), the efficiency for the slope is $e(4) \approx 0.195$.

Discriminative vs. Generative Models



Discriminative vs. Generative Models

- ▶ Next, suppose that the true underlying data-generating mechanism follows:

$$Y \sim \text{Ber}(0.5), \quad X \mid Y = 0 \sim -2 + t(2), \quad \text{and} \quad X \mid Y = 1 \sim 2 + t(2). \quad (28)$$

Here, $t(2)$ denotes a Student's t -distribution with 2 degrees of freedom, whose PDF is given by:

$$f(x) = \frac{1}{2\sqrt{2}} \left(1 + \frac{x^2}{2} \right)^{-3/2}. \quad (29)$$

Due to this structural form, it exhibits a much higher probability of observing extreme values (heavy tails) than a Gaussian distribution.

- ▶ We then investigate the consequences of naively applying both logistic regression and LDA, completely ignoring this model misspecification.

Discriminative vs. Generative Models

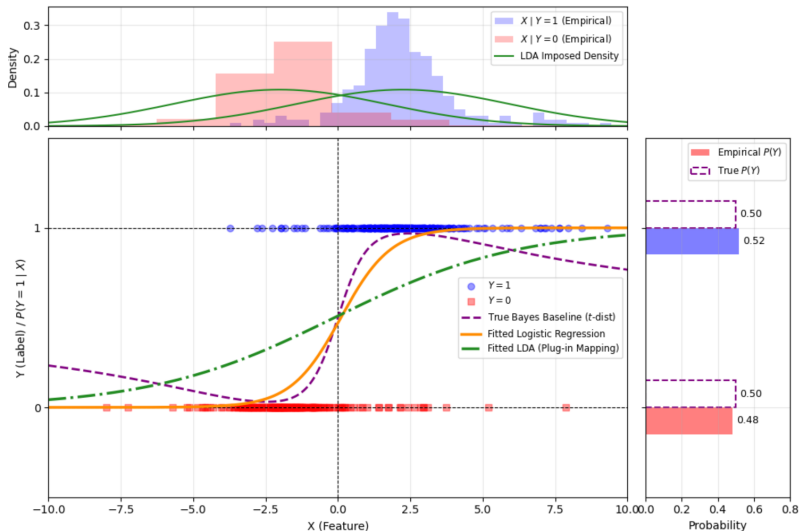
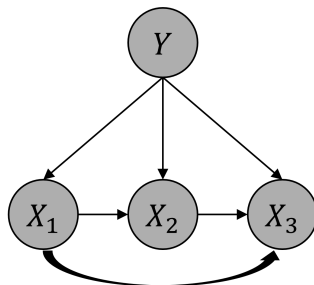


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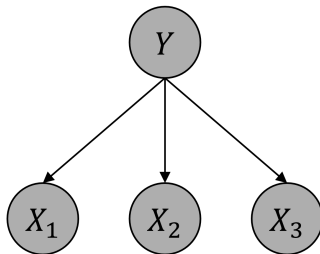
Multivariate Generative Model



- ▶ Multivariate data often exhibits complex dependency structures.
- ▶ The generative model requires learning $f_{\theta}(\mathbf{x} \mid y)$. For example, when $d = 3$:

$$\begin{aligned} f_{\theta}(\mathbf{x}, y) &= p_{\theta}(y) f_{\theta}(\mathbf{x} \mid y) \\ &= p_{\theta}(y) \left(f_{\theta}(x_1 \mid y) f_{\theta}(x_2 \mid y, x_1) f_{\theta}(x_3 \mid y, x_1, x_2) \right). \end{aligned} \tag{30}$$

Multivariate Generative Model: Naive Bayes



- ▶ *Naive Bayes* is a multivariate generative model assuming complete conditional independence:

$$f_{\theta}(\mathbf{x} \mid y) = \prod_{j=1}^d f_{\theta}(x_j \mid y). \quad (31)$$

Multivariate Generative Model: Naive Bayes

- ▶ The assumption of conditional independence simplifies the logit structure:

$$\begin{aligned}\eta_{\theta}(\mathbf{x}) &= \log \frac{P_{\theta}(Y = 1 \mid \mathbf{X} = \mathbf{x})}{P_{\theta}(Y = 0 \mid \mathbf{X} = \mathbf{x})} \\ &= \log \left(\frac{f_{\theta}(\mathbf{x} \mid y = 1)p_{\theta}(y = 1)}{f_{\theta}(\mathbf{x} \mid y = 0)p_{\theta}(y = 0)} \right) \\ &= \sum_{j=1}^d \log \frac{f_{\theta}(x_j \mid y = 1)}{f_{\theta}(x_j \mid y = 0)} + \log \frac{p_{\theta}(y = 1)}{p_{\theta}(y = 0)}.\end{aligned}\tag{32}$$

- ▶ The problem of learning $\log \frac{f_{\theta}(\mathbf{x}|y=1)}{f_{\theta}(\mathbf{x}|y=0)}$ reduces to learning d univariate logits.
- ▶ For example, if $X_j \mid Y = y \sim \mathcal{N}(\mu_{yj}, \sigma_j^2)$, the logit becomes linear:

$$\eta_{\theta}(\mathbf{x}) = \sum_{j=1}^d \left(\frac{\mu_{1j} - \mu_{0j}}{\sigma_j^2} x_j + \frac{\mu_{0j}^2 - \mu_{1j}^2}{2\sigma_j^2} \right) + \log \frac{p_{\theta}(y = 1)}{p_{\theta}(y = 0)}.\tag{33}$$

Multivariate Generative Model: Multivariate LDA

- ▶ Unlike Naive Bayes, multivariate LDA models the full dependency structure:

$$\mathbf{X} \mid Y = y \sim \mathcal{N}(\boldsymbol{\mu}_y, \Sigma). \quad (34)$$

- ▶ The covariance matrix Σ captures dependencies between features and is shared across all classes.
- ▶ The conditional PDF is given by:

$$f_{\theta}(\mathbf{x} \mid y) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_y)^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_y) \right). \quad (35)$$

where $\theta = (\boldsymbol{\mu}_0^T, \boldsymbol{\mu}_1^T, \text{vec}(\Sigma))^T$ is the parameter set containing $O(d^2)$ parameters.

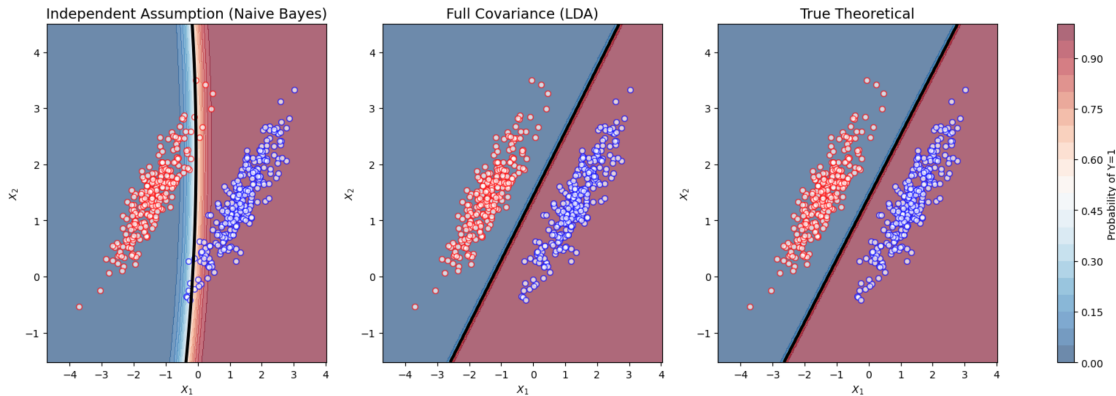
Multivariate Generative Model: Multivariate LDA

- ▶ The logit $\eta_{\theta}(\mathbf{x})$ can be expressed as:

$$\begin{aligned}\eta_{\theta}(\mathbf{x}) &= \log \frac{f_{\theta}(\mathbf{x} \mid y = 1)p_{\theta}(y = 1)}{f_{\theta}(\mathbf{x} \mid y = 0)p_{\theta}(y = 0)} \\ &= (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T \boldsymbol{\Sigma}^{-1} \left(\mathbf{x} - \frac{\boldsymbol{\mu}_1 + \boldsymbol{\mu}_0}{2} \right) + \log \frac{p_{\theta}(y = 1)}{p_{\theta}(y = 0)}.\end{aligned}\tag{36}$$

- ▶ The logit remains linear. However, $\boldsymbol{\Sigma}^{-1}$ explicitly accounts for cross-feature dependencies when defining the weight vector $\mathbf{w} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$.
- ▶ When $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}_d$, multivariate LDA reduces to Gaussian naive Bayes.

Multivariate Generative Model



Curse of Dimensionality

- ▶ As d grows much faster than the sample size n ($d/n \rightarrow \infty$), multivariate LDA performs no better than random guessing (Bickel and Levina, 2004).
- ▶ When $P(Y = 1) = 0.5$, the plug-in classification rule based on multivariate LDA is formulated as:

$$\hat{\delta}_{\text{LDA}}(\mathbf{X}) = I \left((\hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_0)^T \hat{\boldsymbol{\Sigma}}^{-1} \left(\mathbf{X} - \frac{\hat{\boldsymbol{\mu}}_1 + \hat{\boldsymbol{\mu}}_0}{2} \right) > 0 \right) \quad (37)$$

where $\hat{\boldsymbol{\mu}}_y$ and $\hat{\boldsymbol{\Sigma}}$ are the MLEs.

- ▶ Note that a random guessing classifier defined as:

$$\delta_{\text{Random}}(\mathbf{X}) = I(U < 0.5), \quad \text{where } U \sim \mathcal{U}(0, 1), \quad (38)$$

yields a misclassification error rate of $P_{\theta}(\delta_{\text{Random}}(\mathbf{X}) \neq Y) = 0.5$.

Curse of Dimensionality

- ▶ The misclassification error rate of multivariate LDA satisfies:

$$P_{\theta}(\hat{\delta}_{\text{LDA}}(\mathbf{X}) \neq Y) \rightarrow 0.5 \quad (39)$$

when $d/n \rightarrow \infty$, even if the true covariance matrix Σ is well-conditioned with bounded eigenvalues (e.g., $\Sigma = I_d$) (Bickel and Levina, 2004).

- ▶ The root cause of this curse of dimensionality is the excessive number of parameters to estimate.
- ▶ When $d \gg n$, the sample covariance matrix $\hat{\Sigma}$ fails to estimate the $O(d^2)$ elements of Σ . Specifically, $\hat{\Sigma}$ becomes singular, making it challenging to compute $\hat{\Sigma}^{-1}$.

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2. Curse of Dimensionality
3. Dimensionality Reduction

Linear Discriminant Analysis

- ▶ *Dimension reduction* is the process of identifying low-dimensional features that capture the essential information in the data and transforming the data accordingly.
- ▶ For example, the LDA classification decision,

$$\hat{\delta}_{\text{LDA}}(\mathbf{X}) = I \left((\hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_0)^T \hat{\boldsymbol{\Sigma}}^{-1} \left(\mathbf{X} - \frac{\hat{\boldsymbol{\mu}}_1 + \hat{\boldsymbol{\mu}}_0}{2} \right) > 0 \right), \quad (40)$$

is based on the scalar projection:

$$\underbrace{\left(\hat{\boldsymbol{\Sigma}}^{-1} (\hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_0) \right)^T}_{=:\mathbf{w}^T} \left(\mathbf{X} - \frac{\hat{\boldsymbol{\mu}}_1 + \hat{\boldsymbol{\mu}}_0}{2} \right). \quad (41)$$

- ▶ In this context, LDA can be viewed as a supervised dimension reduction method that utilizes the class label Y .

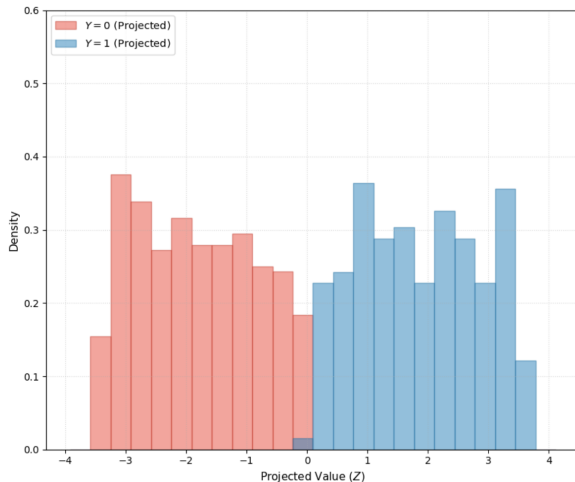
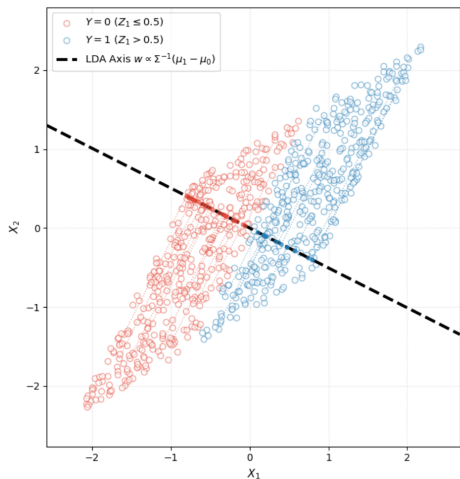
Linear Discriminant Analysis

- ▶ Even when the conditional Gaussian assumption is violated, the LDA projection axis can still be justified as the solution to:

$$\hat{\Sigma}^{-1}(\hat{\mu}_1 - \hat{\mu}_0) \propto \arg \max_{\mathbf{w}: \|\mathbf{w}\|=1} \left| \widehat{\text{Var}}^{-1/2}[\mathbf{w}^T \mathbf{X}] \left(\hat{\mathbb{E}}[\mathbf{w}^T \mathbf{X} \mid Y=1] - \hat{\mathbb{E}}[\mathbf{w}^T \mathbf{X} \mid Y=0] \right) \right|. \quad (42)$$

- ▶ That is, the LDA projection axis maximizes the Mahalanobis distance between the two class centers.

Linear Discriminant Analysis



Principal Component Analysis

- ▶ While LDA effectively identifies the axis that discriminates between the two classes, labeled data are typically scarce in practice.
- ▶ Principal Component Analysis (PCA) is a popular unsupervised dimension reduction method that does not utilize class labels.
- ▶ PCA first finds the direction preserving the variability in the data as much as possible:

$$\mathbf{w}_1 = \arg \max_{\mathbf{w}_1: \|\mathbf{w}_1\|=1} \widehat{\text{Var}}[\mathbf{w}_1^T \mathbf{X}] \quad (43)$$

- ▶ Consequently, the r -th principal axis is sequentially obtained by maximizing the remaining variance, subject to being orthogonal to all previously defined axes:

$$\mathbf{w}_j = \arg \max_{\substack{\mathbf{w}_j: \|\mathbf{w}_j\|=1, \\ \mathbf{w}_j^T \mathbf{w}_k=0, \forall k < j}} \widehat{\text{Var}}[\mathbf{w}_j^T \mathbf{X}] \quad (44)$$

Principal Component Analysis

- ▶ Since $\widehat{\text{Var}}[\mathbf{w}_j^T \mathbf{X}] = \mathbf{w}_j^T \hat{\Sigma} \mathbf{w}_j$, the sample covariance matrix $\hat{\Sigma}$ fully characterizes all of the principal components (PCs).
- ▶ The PCA axes correspond to the eigenvectors of $\hat{\Sigma}$, where the associated eigenvalues λ_j directly quantify the explained variance along each component:

$$\hat{\Sigma} \mathbf{w}_j = \lambda_j \mathbf{w}_j \quad \text{where} \quad \lambda_j = \widehat{\text{Var}}[\mathbf{w}_j^T \mathbf{X}] \quad (45)$$

where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0$.

- ▶ The total sample variance, representing the entire variability in $\mathbf{X}$, is preserved under this orthogonal transformation:

$$\sum_{j=1}^d \widehat{\text{Var}}[X_j] = \text{tr}(\hat{\Sigma}) = \sum_{j=1}^d \lambda_j = \sum_{j=1}^d \widehat{\text{Var}}[\mathbf{w}_j^T \mathbf{X}] \quad (46)$$

Principal Component Analysis

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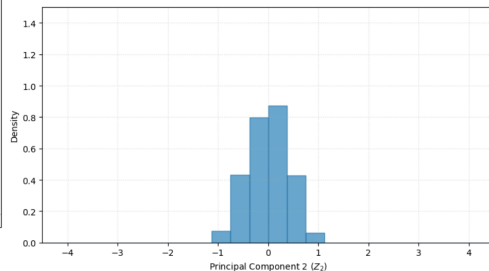
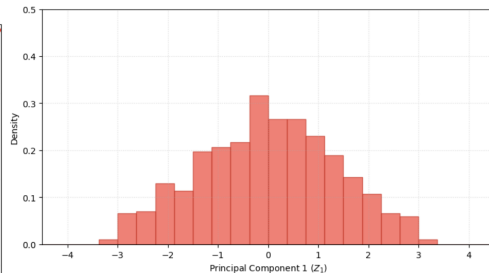
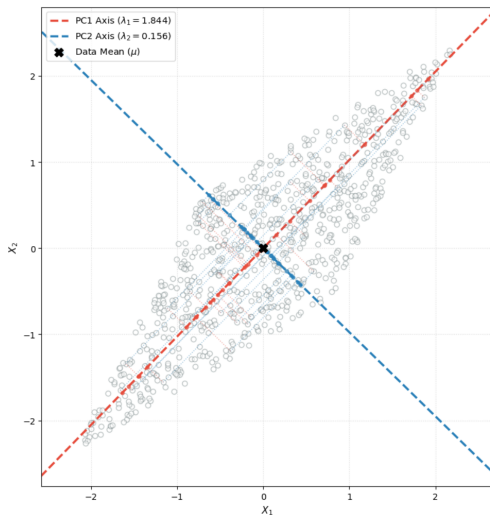
$$\hat{\Sigma} \mathbf{w}_j = \lambda_j \mathbf{w}_j \quad \text{where} \quad \lambda_j = \widehat{\text{Var}}[\mathbf{w}_j^T \mathbf{X}] \quad (47)$$

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Principal Component Analysis



Principal Component Analysis

- ▶ We can reconstruct the data vector $\mathbf{X}$ using the top- k PCs through a mirrored encoding-decoding mapping:

$$\mathbf{X} \xrightarrow{\text{Embedding}} W^T(\mathbf{X} - \hat{\boldsymbol{\mu}}) \xrightarrow{\text{Reconstruction}} \hat{\boldsymbol{\mu}} + WW^T(\mathbf{X} - \hat{\boldsymbol{\mu}}) \quad (49)$$

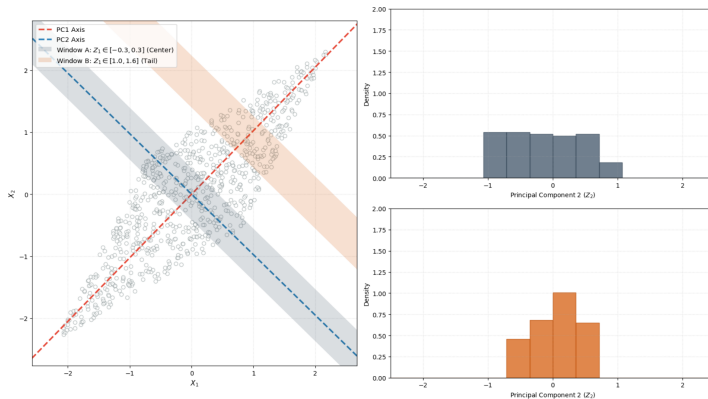
where $W = [\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k]$ represents the $d \times k$ projection matrix.

- ▶ The mean squared error (called *reconstruction error*) under this subspace approximation can be explicitly quantified as:

$$n^{-1} \sum_{i=1}^n \left\| \mathbf{x}_i - \left(\hat{\boldsymbol{\mu}} + WW^T(\mathbf{x}_i - \hat{\boldsymbol{\mu}}) \right) \right\|^2 = \text{tr}(\hat{\boldsymbol{\Sigma}}) - \sum_{j=1}^k \widehat{\text{Var}}[\mathbf{w}_j^T \mathbf{X}] \quad (50)$$

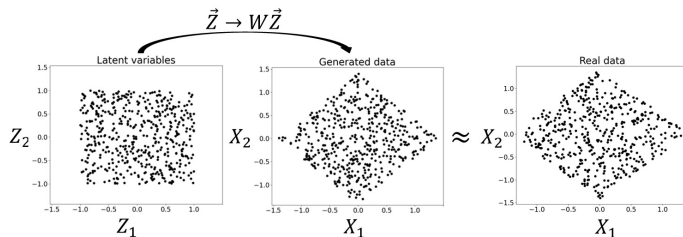
- ▶ Thus, the PCA axes can be interpreted as the minimizers of the reconstruction error under the orthonormality constraint.

Independent Component Analysis



- ▶ Although PCA identifies orthogonal components, orthogonality or uncorrelatedness alone does not imply statistical independence.

Independent Component Analysis



- ▶ Independent Component Analysis (ICA) (Hyvärinen and Oja, 2000) identifies latent sources and utilizes them to achieve statistically independent dimension reduction.
- ▶ ICA assumes a linear generative structure: $\mathbf{X} = W\mathbf{Z}$, where $W \in \mathbb{R}^{d \times k}$ and $W^T W = I_k$. Here, $\mathbf{Z}$ denotes latent variables following a known non-Gaussian distribution (e.g., $\mathcal{U}([0, 1]^2)$ or $\mathcal{U}([-1, 1]^2)$).
- ▶ Assuming all underlying sources are mutually independent, the latent variables can be explicitly inferred via $\mathbf{Z} = W^T \mathbf{X}$.

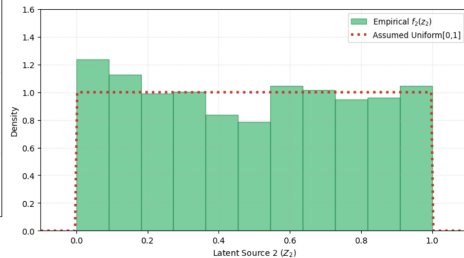
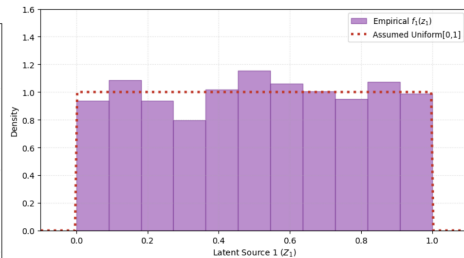
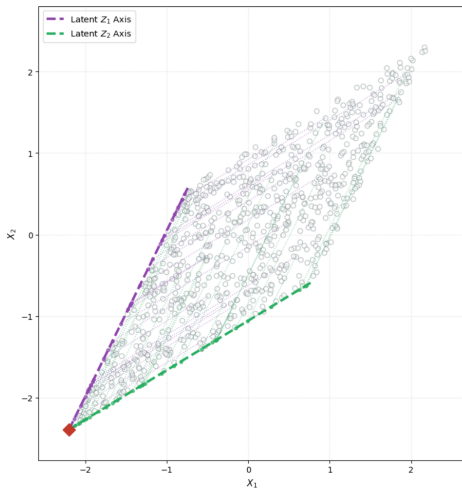
Independent Component Analysis

- ▶ There are several criteria for learning this linear generative structure, including kurtosis maximization, mutual information minimization, and maximum likelihood estimation.
- ▶ Focusing on the likelihood-maximization approach,³ the change of variables formula yields $p_{\theta}(\mathbf{x}) = p_{\mathbf{Z}}(\mathbf{z} = W^T \mathbf{x}) \cdot |\det W^T|$.
- ▶ Since W is orthogonal ($|\det W^T| = 1$), the log-likelihood $l_n(\theta)$ simplifies via strict statistical independence to:

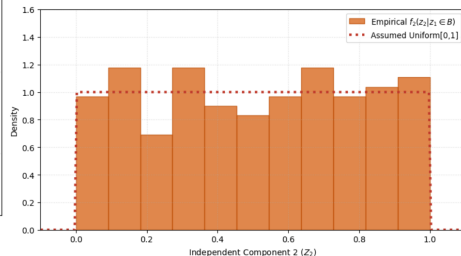
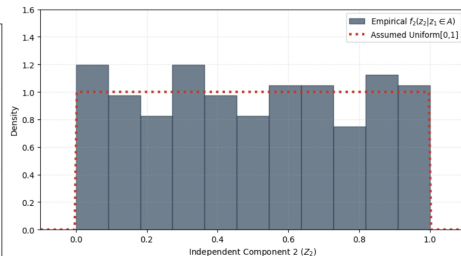
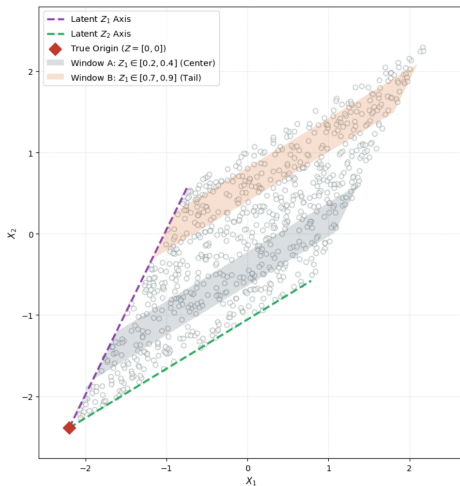
$$l_n(\theta) = \sum_{i=1}^n \log p_{\mathbf{Z}}(W^T \mathbf{x}_i) = \sum_{i=1}^n \sum_{j=1}^d \log p_{Z_j} \left((W^T \mathbf{x}_i)_j \right) \quad (51)$$

³Although ICA is generally applicable when $k < d$, for simplicity, we assume $k = d$ so that $W^T = W^{-1}$.

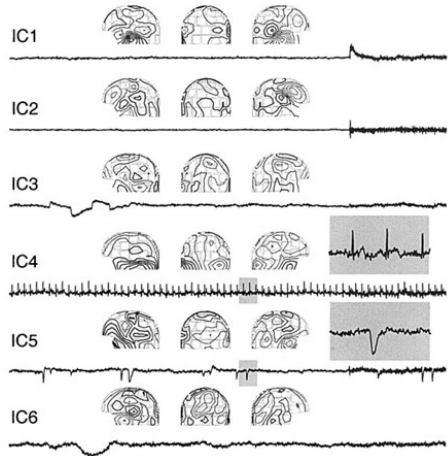
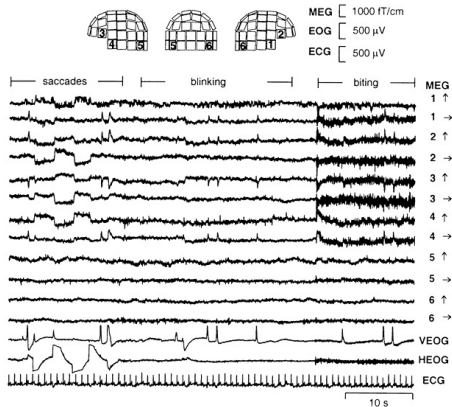
Independent Component Analysis



Independent Component Analysis

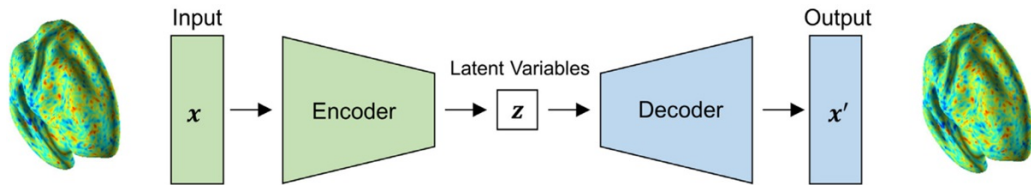


Independent Component Analysis



Figures are edited from Hyvärinen and Oja (2000).

Nonlinear Methods: Autoencoders



- ▶ As a non-linear extension of PCA, we can generalize the framework using an autoencoder (Kramer, 1991; Plaut, 2018).
- ▶ Autoencoders (AEs) (Bengio et al., 2006) consist of pairs of encoders and decoders that efficiently reduce the dimensionality of data.
- ▶ The prefix 'auto' is used because they autonomously learn to encode data in an unsupervised manner.

Images are from Kim et al. (2021).

Nonlinear Methods: Autoencoders

- Specifically, we may consider a flexible encoding function $f_{\text{enc}}(\cdot)$ and a linear decoder W :

$$\mathbf{X} \xrightarrow{\text{Embedding}} f_{\text{enc}}(\mathbf{X}) \xrightarrow{\text{Reconstruction}} \mathbf{b} + Wf_{\text{enc}}(\mathbf{X}) \quad (52)$$

- To understand its foundation, consider minimizing the reconstruction error over all possible encoding results $f_{\text{enc}}(\mathbf{x}_1), \dots, f_{\text{enc}}(\mathbf{x}_n) \in \mathbb{R}^k$ subject to $W^T W = I_k$:

$$n^{-1} \sum_{i=1}^n \|\mathbf{x}_i - (\mathbf{b} + Wf_{\text{enc}}(\mathbf{x}_i))\|^2 \geq \text{tr}(\hat{\Sigma}) - \sum_{j=1}^k \widehat{\text{Var}}[\mathbf{w}_j^T \mathbf{X}] \quad (53)$$

- It is known that the lower bound is achieved if and only if:

$$\mathbf{b} + Wf_{\text{enc}}(\mathbf{x}_i) = \hat{\boldsymbol{\mu}} + W^T(\mathbf{x}_i - \hat{\boldsymbol{\mu}})$$

This implies that the first k PCs are the optimal choices.

- See Section 14.5.1 in Hastie (2009) for further theoretical details.

Nonlinear Methods: Variational Autoencoders

- ▶ Variational autoencoders (VAEs) (Kingma and Welling, 2014) can be understood as a non-linear extension of Independent Component Analysis (ICA) (Khemakhem et al., 2020).
- ▶ A more rigorous and detailed discussion on variational autoencoders will be presented in a later chapter.

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